~~But~~ $$\boxed{\frac{dw}{dt} = \frac{dw}{dx}\frac{dx}{dt} + \frac{dw}{dy}\frac{dy}{dt}}$$

But in Eqn: 3

- w depends on three variables, x, y, z.
- x, y, z depend on two variables r and s.

⇒ 14.5

Directional Derivatives and Gradient vector.

⇒ Using the defination, find the derivative of

from Book:

you have a function

$$f(x,y) = x^2 + xy$$

you are at the point

$$P_0\ (1,2)$$

you want the derivative in the direction of

$$U = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

and we must use the formula:

$$\left(\frac{df}{ds}\right)_{U,P_0} = \lim_{s \to 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

- $x_0 = 1$
- $y_0 = 2$

$u_1 = \frac{1}{\sqrt{2}}$

$u_2 = \frac{1}{\sqrt{2}}$

So we plug in:

$$\frac{df}{ds} = \lim_{s \to 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}$$

$$f\left(1 + s\frac{1}{\sqrt{2}}, 2 + s\frac{1}{\sqrt{2}}\right) - f(1,2)$$

Step 2. Plug into the function:

$$f(x,y) = x^2 + xy$$

$$f\left(1 + \frac{s}{\sqrt{2}}, 2 + \frac{s}{\sqrt{2}}\right)$$

This becomes

$$\left(1 + \frac{s}{\sqrt{2}}\right)^2 + \left(1 + \frac{s}{\sqrt{2}}\right)\left(2 + \frac{s}{\sqrt{2}}\right)$$

$$f(1,2) = 1^2 + 1 \cdot 2 = 3$$

Expand the squares:

First part:

$$\left(1 + \frac{s}{\sqrt{2}}\right)^2 = 1 + \frac{2s}{\sqrt{2}} + \frac{s^2}{2}$$

Second part:

$$\left(1 + \frac{s}{\sqrt{2}}\right)\left(2 + \frac{s}{\sqrt{2}}\right)$$

Multiply term by term,

$$= 2 + \frac{3s}{\sqrt{2}} + \frac{s^2}{2}$$

Step 4. Add both parts and sub 3.

$$\left(1 + \frac{2s}{\sqrt{2}} + \frac{s^2}{2}\right) + \left(2 + \frac{3s}{\sqrt{2}} + \frac{s^2}{2}\right)$$

$$-3+\frac{5s}{\sqrt{2}}+s^2-3$$

$$=\frac{5s}{\sqrt{2}}+s^2$$

steps, Plug into the limit formula:-

$$\frac{df}{ds}=\lim_{s\to 0}\frac{\frac{5s}{\sqrt{2}}+s^2}{s}$$

$$=\lim_{s\to 0}\left(\frac{5}{\sqrt{2}}+s\right)$$

step6, Take the limit:-

$$\frac{5}{\sqrt{2}}+s\to\boxed{\frac{5}{\sqrt{2}}\text{ Ans}}$$

again using book.

⇒ Using the defination, find the derivative of $f(x,y)=x^2+xy$ at P_0 $(1,2)$ in the direction of the unit vector $u=\left(\frac{1}{\sqrt{2}}\right)i+\left(\frac{1}{\sqrt{2}}\right)j$

sol, Applying the def in eq (1)

$$\left(\frac{df}{ds}\right)_{u,P_0}=\lim_{s\to 0}\frac{f(x_0+su_1,\ y_0+su_2)-f(x_0,y_0)}{s}\quad\text{eq (i)}$$

$$\lim_{s\to 0}\frac{\left(1+s\left(\frac{1}{\sqrt{2}}\right),\ 2+s\left(\frac{1}{\sqrt{2}}\right)-f(1,2)\right)}{s}$$

$$\lim_{s\to 0} \frac{\left(1+\frac{s}{\sqrt{2}}\right)^2 + \left(1+\frac{s}{\sqrt{2}}\right)\left(2+\frac{s}{\sqrt{2}}\right) - f(1+1.2)}{s}$$

$$= \lim_{s\to 0} \frac{\left(1+\frac{2s}{\sqrt{2}}+\frac{s^2}{2}\right)+\left(2+\frac{3s}{\sqrt{2}}+\frac{s^2}{2}\right)-3}{s}$$

$$= \lim_{s\to 0} \frac{\frac{5s}{\sqrt{2}}+s^2}{s} = \lim_{s\to 0}\left(\frac{5}{\sqrt{2}}+s\right) = \frac{5}{\sqrt{2}}$$

The rate of change of $f(x,y) = x^2+xy$ at P_0 $(1,2)$ in the direction U is $5/\sqrt{2}$

Ans

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e.g. Theory: The formula for directional derivative, $D_u f$, is.

$$D_u f(x_0, y_0) = \nabla f(x_0, y_0) \cdot u$$